

MCA SEMESTER-I
School of Computer and Systems Sciences,
Jawaharlal Nehru University,
New Delhi -110067

Computer Fundamental (CS-402)

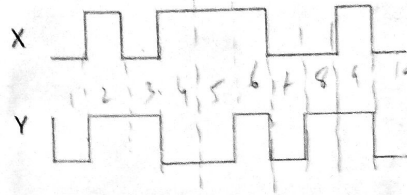
Date: 02.03.2023

Total Marks: 50

Duration: 3hr

All Compulsory

1. Perform the a) OR b) AND c) EX-OR d) EX-NOR e) NOT for following pulse waveforms: 5



2. Prove the theorems: a) $x+x=x$ b) $x.x=x$ c) $x+1=1$ d) $x+xy=x$ 5

3. Simplify the Boolean Function F using don't care condition 'd' : 5

$$F = A'B'D' + A'CD + A'BC$$

$$d = A'BC'D + ACD + AB'D'$$

Implement the simplified function using 2-input gates.

4. Implement a 3x8 decoder. 5

5. Implement a full adder using 2-input gates. 5

6. Implement the following function with a 8X1 multiplexer:
 $F(A,B,C,D) = \Sigma(0,1,3,4,8,9,15)$ 5

7. Implement a 3-bit magnitude comparator. 7

8. Implement a 8x4 ROM for the following: 7

x	y	z	F1	F2	F3	F4
0	0	0	0	1	1	0
0	0	1	0	1	0	0
0	1	0	0	0	1	0
0	1	1	0	0	0	1
1	0	0	1	1	0	0
1	0	1	1	1	1	0
1	1	0	1	0	0	0
1	1	1	1	0	0	1

9. Implement the following functions with PLA: 6

$$F_1(A, B, C) = \Sigma(3, 5, 6, 7)$$

$$F_2(A, B, C) = \Sigma(0, 2, 4, 7)$$

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End sem-I

Date: 20.12.2024@10am

Total Marks: 50

Time: 3hr

All Compulsory

1. Write note on
 - a. Parity checker and generator
 - b. The need of minimization in a logic circuit
 - c. Combinational and sequential circuits
 - d. Any two ID digital logic families
 - e. Edge triggering and Level triggering

3*5=15

2. Represent decimal number 8620 in
 - a. BCD
 - b. Excess-3 code
 - c. 2421
 - d. Binary
 - e. Hexadecimal

5

3. What will be the output signal f and g for logic circuit shown in fig 1 with following a, b and c input signals shown in Fig 2?

5

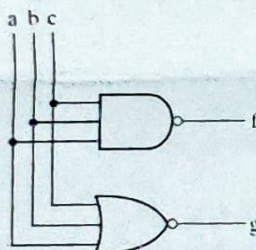


Fig 1 : Logic Circuit

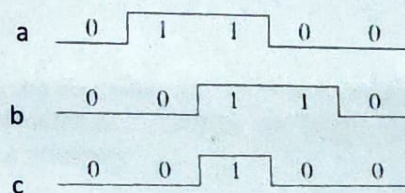


Fig 2 : Input Signal

4. Simplify the Boolean Function F together with don't care conditions d in (i) sum of product and (ii) Product of sum

$$F(w, x, y, z) = \Sigma(0, 1, 2, 3, 7, 8, 10)$$

$$d(w, x, y, z) = \Sigma(5, 6, 11, 15)$$

5

5. Implement a 3 bit magnitude comparator.

5

6. Implement the following functions with PLA

$$F_1(A, B, C) = \Sigma(3, 5, 6, 7)$$

$$F_2(A, B, C) = \Sigma(0, 2, 4, 7)$$

5

7. a) What is a flip flop? Explain the T-flip flop alongwith state table, characteristic table with characteristic equation.
 b) Implement 3 bit step-up counter using T flip. Make state diagram, excitation table and logic circuit.

4+6

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End Semester Exam

Total Marks: 50

Date: 08.12.2025

Duration: 3hr

All Compulsory

1. Write note on:
 - a. Basic building block of the logic Gate
 - b. Status Register
 - c. Shift register.
 - d. Accumulator
 - e. Hardwired control and microprogram sequencer control

3x5=15
2. For the following Boolean functions solve accordingly.
 - a. Express the Boolean function $F = xy + x'z$ in a product of sum
 - b. Obtain the truth table of the function F and express each in sum of minterms and product of maxterms. $F = (A' + B)(B' + C)$

2+3=5
3. Simplify the Boolean function F together with don't care condition d , then express the simplified function in sum of minterms.
 $F(A, B, C, D) = \Sigma(1, 3, 5, 7, 9, 15)$
 $d(A, B, C, D) = \Sigma(4, 6, 12, 13)$

5
4. Design and implement full Adder.

5
5. Implement function using multiplexer
 $F(A, B, C, D) = \Sigma(0, 1, 3, 4, 8, 9, 15)$

5
6. Combinational circuit is defined by the function :
 $F_1(A, B, C) = \Sigma(3, 5, 6, 7)$
 $F_2(A, B, C) = \Sigma(0, 2, 4, 7)$
Implement the circuit using PLA.

5
7. a) Explain T-flip flop with logic diagram and characteristic table.
b) Implement 3-bit binary counter using T-Flip flop.

5+5=10